Trade Tariffs and Exchange Rates in a Multilateral World

Theoretical derivations

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1 Introduction

In a two-country model, an import tariff appreciates the tariffing country's currency under the Marshall–Lerner condition. In a multilateral setting, a discriminatory tariff also reallocates expenditure across foreign suppliers, so bilateral exchange-rate responses need not share a sign across countries. These notes characterize the exchange-rate responses in an N-country model. The general equilibrium response is $(-J)^{-1}F$, where F is the vector of trade-balance disturbances created by the tariff at unchanged exchange rates and J is the exchange-rate adjustment mechanism. Under symmetry and a multilateral Marshall–Lerner condition, the tariffing country always appreciates against the target, but the sign of its bilateral exchange-rate response against an untariffed bystander is ambiguous and depends on whether the cross-origin substitution elasticity exceeds a threshold $N\rho_1^*$. Despite this bilateral sign ambiguity, the tariffing country's nominal effective exchange rate appreciates independently of the cross-origin elasticity and N. The exact cancellation behind this aggregate result and the linear threshold are properties of the symmetric nested-CES model, but the disturbance-adjustment decomposition, and the distinction between aggregate home-versus-foreign adjustment and reallocation across foreign suppliers, are more general. Proofs of all formal results are collected in Appendix D.

2 N-Country Model and Equilibrium

2.1 Production and preferences

Countries are indexed by $i, j \in \{1, \ldots, N\}$; generically, A denotes the tariff imposer, B the tariff target, and C an untariffed bystander. Each country produces a country-specific tradable and a non-tradable from labor:

$$Y_{T_i} = A_{T_i} L_{T_i}, \qquad Y_{N_i} = A_{N_i} L_{N_i}, \qquad L_{T_i} + L_{N_i} = L_i, \tag{1}$$

with perfect competition and producer-price normalization $P_{T_i} = 1$ in i's currency, so

$$w_i = A_{T_i}, \qquad P_{N_i} = A_{T_i}/A_{N_i}. \tag{2}$$

Wages are constant in own currency, so all relative adjustment runs through exchange rates. Preferences are nested CES with share-linear weights:

$$C_i = C_{T_i}^{\alpha_T}\, C_{N_i}^{1-\alpha_T} \qquad \text{(tradable share } \alpha_T\text{)} \tag{3}$$

$$C_{T_i} = \left[\alpha_D^{1/\eta}\, C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta}\, M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}} \qquad \text{(home weight } \alpha_D,\ \text{elasticity } \eta\text{)} \tag{4}$$

$$M_i = \left[\ \textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho}\, C_{T_j i}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \qquad \text{(origin weights } \beta_{ij},\ \text{elasticity } \rho\text{).} \tag{5}$$

η governs substitution between domestic and imported tradables (the home-versus-foreign tradables margin); ρ governs substitution across foreign origins conditional on importing (the foreign-supplier tradables reallocation margin, inactive at $N = 2$ and active at $N \geq 3$).

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N-1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Home bias α_D keeps home and foreign tradables asymmetric; full home-versus-foreign tradables symmetry would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, and is not imposed.

2.2 Tariffs, prices, income, and trade balance

Tariffs $\tau_{ij} \geq 0$ are ad valorem, levied by i on imports from j, with $\tau_{ii} = 0$. Consumer prices in i's currency are given by $p_{ij} = (1+\tau_{ij})\, e_{ij}$, $p_{ii} = 1$, and price indices are given by:

$$P_{M_i} = \left[\textstyle\sum_{j\neq i} \beta_{ij}\, p_{ij}^{1-\rho}\right]^{\frac{1}{1-\rho}}, \qquad P_{T_i} = \left[\alpha_D\, p_{ii}^{1-\eta} + (1-\alpha_D)\, P_{M_i}^{1-\eta}\right]^{\frac{1}{1-\eta}}, \qquad P_i = \kappa_i\, P_{T_i}^{\alpha_T}\, P_{N_i}^{1-\alpha_T}. \tag{6}$$

Expenditure shares of income are given by:

$$s_{ii} = \alpha_T\, \alpha_D \left(\frac{p_{ii}}{P_{T_i}}\right)^{1-\eta}, \qquad s_{ij} = \alpha_T\,(1-\alpha_D)\left(\frac{P_{M_i}}{P_{T_i}}\right)^{1-\eta} \beta_{ij}\left(\frac{p_{ij}}{P_{M_i}}\right)^{1-\rho}, \quad j \neq i, \tag{7}$$

therefore, $s_{ij} I_i$ is tariff-inclusive spending at consumer prices, $s_{ij} I_i/(1+\tau_{ij})$ is the tariff-exclusive border value. Revenue is rebated lump-sum,

$$I_i = w_i L_i + \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij} I_i \quad \Longrightarrow \quad I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij}}. \tag{8}$$

Trade balances at producer prices in a common currency (since the tariff wedge is a domestic transfer):

$$\mathrm{TB}_i = \sum_{k\neq i} f_{ki} - \sum_{j\neq i} f_{ij}, \qquad f_{ij} \equiv \frac{s_{ij}\, I_i}{1+\tau_{ij}}. \tag{9}$$

By Walras' law only $N-1$ conditions are independent. Furthermore, we impose closure, i.e., no internationally traded assets: $\mathrm{TB}_i = 0$ for all i. Therefore, given $\{\tau_{ij}\}$, the $N-1$ free rates solve $\mathrm{TB}_i = 0$, $i \neq A$, in equilibrium.

2.3 Exchange-rate concepts

Define e_{ij} = units of i's currency per unit of j's, $e_{ii} = 1$, $e_{ij} = e_{ik}e_{kj}$. Therefore, a rise in e_{ij} is a depreciation of i against j. Log perturbations in bilateral exchange rates relative to A are defined by $\nu_j \equiv d\log e_{Aj}$, $\nu_A = 0$. Define relative wages, terms of trade, real exchange rates, and effective exchange rates:

$$\omega_{ij} \equiv \frac{e_{ij}w_j}{w_i}, \qquad \mathrm{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij}P_{T_j}} = \frac{1}{e_{ij}}, \qquad q_{ij} \equiv \frac{e_{ij}P_j}{P_i}, \tag{10}$$

$$d\nu_i^E \equiv \sum_{j\neq i} w_{ij}\, d\log e_{ij}, \qquad dq_i^E \equiv \sum_{j\neq i} w_{ij}\, d\log q_{ij}, \qquad w_{ij} = \frac{f_{ij}+f_{ji}}{\sum_{k\neq i}(f_{ik}+f_{ki})}, \tag{11}$$

with $w_{ij} = 1/(N-1)$ in the symmetric model. Under the production normalization,

$$d\log e_{ij} = d\log \omega_{ij} = -\, d\log \mathrm{ToT}_{ij}, \tag{12}$$

so every nominal exchange-rate result is equivalently a relative-wage or terms-of-trade result.

3 General Results on Exchange-Rate Adjustment

Exchange-rate determination under the trade-balance closure has a structure that does not depend on the nested-CES specification.

3.1 The local disturbance–adjustment representation

Consider any differentiable multilateral trade environment with $N-1$ independent trade-balance conditions, $\mathrm{TB}_i = 0$ for $i \neq A$, as functions of the exchange rates and the tariffs. Stacking $d\nu \equiv (\nu_j)_{j\neq A}$ and the tariff perturbations $d\boldsymbol{\tau} \equiv (dt_{jk})_{j\neq k}$, the conditions for $i \neq A$ are

$$J\, d\nu + G\, d\boldsymbol{\tau} = 0, \qquad J_{il} \equiv \frac{\partial\, d\mathrm{TB}_i}{\partial \nu_l}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial\, d\mathrm{TB}_i}{\partial\, dt_{jk}}\right|_{\nu \text{ fixed}}. \tag{13}$$

Separating the tariff perturbation into a direction a scaled by $d\tau$ (unilateral tariff: $a_{AB} = 1$; war: $a_{AB} = a_{BA} = 1$) gives the impact vector

$$F \equiv Ga \qquad \Longrightarrow \qquad J\, d\nu = -F\, d\tau, \qquad \frac{d\nu}{d\tau} = (-J)^{-1}F. \tag{14}$$

F is the vector of trade-balance disturbances created by the tariff at unchanged exchange rates (incorporating multilateral substitution and income effects), J describes how trade balances respond to exchange rates, and configurations differ only in the direction a applied to the same G.

3.2 The multilateral Marshall–Lerner condition

Let $\widetilde{J}$ be the full $N \times N$ Jacobian before dropping country A. Homogeneity (trade balances depend only on relative currency values) and Walras' law give the identities

$$\sum_l \widetilde{J}_{il} = 0 \quad \forall i, \qquad\qquad \sum_i \widetilde{J}_{il} = 0 \quad \forall l. \tag{15}$$

Assumption 1 (Gross substitutability)**.** *$\widetilde{J}_{il} \geq 0$ for all $l \neq i$. In the symmetric systems below every off-diagonal entry is a positive multiple of D_N, so the assumption there is $D_N > 0$, which holds for $\eta \geq 1$.*

Homogeneity and Assumption 1 give $\widetilde{J}_{ii} = -\sum_{l \neq i} \widetilde{J}_{il} \leq 0$ (i.e., the bilateral Marshall–Lerner conditions), and the reduced J inherits a dominant diagonal,

$$|J_{ii}| = \sum_{l \neq i,\, A} J_{il} + \widetilde{J}_{iA} \;\geq\; \sum_{l \neq i,\, A} J_{il}, \tag{16}$$

strict in every row with $\widetilde{J}_{iA} > 0$. By the standard characterization of irreducibly diagonally dominant Z-matrices:

Proposition 3.1 (Multilateral Marshall–Lerner condition)**.** *Under Assumption 1, with J irreducible and $\widetilde{J}_{iA} > 0$ for some i, $-J$ is a nonsingular M-matrix:*

$$\det(-J) > 0, \qquad (-J)^{-1} \geq 0 \textit{ elementwise}, \qquad \operatorname{Re} \lambda(J) < 0. \tag{17}$$

In words, the multilateral Marshall–Lerner condition generalizes the bilateral Marshall–Lerner condition to require that the exchange-rate adjustment mechanism is regular in the sense that each country's own depreciation improves its own trade balance, no country's appreciation worsens another country's balance, and the own effect dominates the cross effects. Therefore, any pattern of trade imbalances is eliminated by "conventional" exchange rate adjustment: surplus countries appreciate, deficit countries depreciate.[1]

Corollary 3.2 (Sign rule)**.** *Under* (17)*,*

$$\operatorname{sign}\!\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\!\Big(\sum_j \vartheta_{ij} F_j\Big), \qquad \vartheta_{ij} \equiv \big[(-J)^{-1}\big]_{ij} \geq 0. \tag{18}$$

Under the multilateral Marshall–Lerner condition, opposite-signed bilateral responses require a mixed-sign F, i.e., bilateral exchange-rate ambiguity comes from the tariff disturbance on trade balances and not from the adjustment mechanism.

[1] Formally: gross substitutability $\Rightarrow$ multilateral Marshall–Lerner. We assume the former as the primitive (Assumption 1), but the subsequent analysis uses only (17). Under the sign structure of Assumption 1, multilateral Marshall–Lerner $\equiv$ local stability of the tatonnement $\dot{\nu}_i = k_i \mathrm{TB}_i$, any $k_i > 0$, which says that simple dynamic exchange-rate adjustment processes converge to the equilibrium.

4 Tariff Responses in the Model

This section computes the entries of J and G for the nested-CES model for $N = 2$, $N = 3$, and general symmetric N.

4.1 The linearized system

Linearizing at a balanced free-trade baseline with normalized incomes, define home share $h_i = \alpha_D$, within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = m\, b_{ij}$ with $m \equiv \alpha_T(1-\alpha_D)$, and flows $f_{ij} = s_{ij} I_i$, $\sum_k f_{ki} = \sum_j f_{ij}$. Perturbing this baseline with tariffs dt_{ij}:

$$dp_{ij} = \nu_j - \nu_i + dt_{ij} \qquad (dp_{ii} = 0), \tag{19}$$

$$d\log P_{M_i} = \sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{T_i} = (1-\alpha_D)\, d\log P_{M_i}, \tag{20}$$

$$d\log s_{ii} = -(1-\eta)(1-\alpha_D)\, d\log P_{M_i}, \tag{21}$$

$$d\log s_{ij} = \underbrace{(1-\rho)\big(dp_{ij} - d\log P_{M_i}\big)}_{\text{across origins}} + \underbrace{(1-\eta)\,\alpha_D\, d\log P_{M_i}}_{\text{home vs. imports}}, \tag{22}$$

$$d\log I_i = \nu_i + \sum_{j\neq i} s_{ij}\, dt_{ij}, \tag{23}$$

$$d\,\mathrm{TB}_i = \sum_{k\neq i} f_{ki}\big(d\log s_{ki} + d\log I_k - dt_{ki}\big) - \sum_{j\neq i} f_{ij}\big(d\log s_{ij} + d\log I_i - dt_{ij}\big). \tag{24}$$

Reading the coefficients off (24) gives the model's entries of J and G in (13), so we obtain specific parametric expressions for every term from the model of Section 2.

The algebraic structure of (14) also bounds how bilateral signs can change with ρ. The elasticity ρ enters the share responses (22) only through the coefficient $(1-\rho)$, so every entry of J and F is linear in ρ. Each entry of $\mathrm{adj}(-J)$ is a determinant of an $(N-2)\times(N-2)$ submatrix of $-J$ and is therefore a polynomial of degree at most $N-2$ in ρ, so each numerator $[\mathrm{adj}(-J)\, F]_i$ in (14) is a polynomial of degree at most $N-1$. Since $\det(-J) > 0$ under (17), a bilateral response can change sign only at a root of its numerator polynomial. At $N = 3$ the symmetric baseline reduces the relevant numerator to a linear function of ρ, giving the single threshold derived below; a general asymmetric baseline gives a quadratic (Appendix B); and at $N \geq 4$ the higher-degree numerators allow a response to change sign more than once as ρ rises.

Now suppose A imposes $d\tau = dt_{AB}$ at the symmetric baseline. Define the following constant, a function of model parameters alone:

$$\rho_1^* \equiv 1 + \alpha_D(\eta-1) - \alpha_T(1-\alpha_D) = \underbrace{\alpha_D\,\eta}_{\text{home switching}} + \underbrace{(1-\alpha_D)(1-\alpha_T)}_{\text{expenditure absorption}} > 0. \tag{25}$$

4.2 Two countries

At $N = 2$ the system (14) is scalar. With $s_{AB} = s_{BA} = m$, equations (19)–(24) collect to

$$d\,\mathrm{TB}_A = \underbrace{m\,D_2\,\nu_B}_{\text{adjustment}} + \underbrace{m\,\rho_1^*\,d\tau}_{\text{impact disturbance}} = 0, \qquad D_2 \equiv 2\alpha_D(\eta - 1) + 1, \tag{26}$$

$$\boxed{\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_1^*}{D_2}.} \tag{27}$$

In reduced form, with export volume $X(1/e)$, import volume $M((1+\tau)e)$, and elasticities $\varepsilon_X, \varepsilon_M$, balanced trade gives $\partial\mathrm{TB}_A/\partial e = M(\varepsilon_X + \varepsilon_M - 1)$; comparing with (26) per unit of import value,

$$\varepsilon_X + \varepsilon_M - 1 = D_2, \qquad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1. \tag{28}$$

Proposition 4.1 (Conventional wisdom, $N = 2$)**.** *At $N = 2$, the multilateral Marshall–Lerner condition (17) is equivalent to $D_2 > 0$, which corresponds exactly to the textbook Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. In our nested-CES model, it holds whenever $\eta \geq 1$. The impact disturbance is $F = m\rho_1^* > 0$ for all admissible parameters, therefore a unilateral import tariff appreciates the tariffing country's currency: $d\log e_{AB}/d\tau = -\rho_1^*/D_2 < 0$.*

Appendix A gives the exact finite-tariff Cobb–Douglas solution.

4.3 Three countries

At $N = 3$ ($\beta_{ij} = \frac{1}{2}$, $s_{ij} = m/2$), (19)–(20) give

$$d\log P_{M_A} = \tfrac{1}{2}(\nu_B + \nu_C + d\tau), \qquad d\log P_{M_B} = \tfrac{1}{2}\nu_C - \nu_B, \qquad d\log P_{M_C} = \tfrac{1}{2}\nu_B - \nu_C, \tag{29}$$

with $d\log I_A = \frac{m}{2}d\tau$, $d\log I_B = \nu_B$, $d\log I_C = \nu_C$. Substituting into (24) for $i = B, C$ and collecting:

$$J = -\frac{mD_3}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \qquad D_3 = 3\alpha_D(\eta - 1) + \rho + 1. \tag{30}$$

The impact trade-balance disturbances are $F_B = -\frac{m}{4}(\rho + \rho_1^*) < 0$ and $F_C = \frac{m}{4}(\rho - \rho_1^*)$. That is, the target B always moves toward deficit, while the bystander gains diverted expenditure (increasing in ρ) against reduced sales to a poorer B and A's home switching, with net gain iff $\rho > \rho_1^*$. Inverting:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1}F\,d\tau = -\frac{1}{3D_3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau = \frac{1}{3D_3}\begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau. \tag{31}$$

Proposition 4.2 (Three-country threshold)**.** *With $D_3 > 0$,*

$$\frac{d\log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D_3} < 0, \qquad \boxed{\frac{d\log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D_3}} \qquad \frac{d\log e_{BC}}{d\tau} = \frac{2\rho}{3D_3} > 0 : \tag{32}$$

A always appreciates against B, B always depreciates against C, and A depreciates against the untariffed C iff $\rho > \rho_3^* \equiv 3\rho_1^*$.

The distinction between the two thresholds deserves emphasis. A favorable trade-diversion effect on C at unchanged exchange rates ($F_C > 0$, i.e. $\rho > \rho_1^*$) is not sufficient for the A–C bilateral rate to reverse. By the sign rule, $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_1^*$: C's equilibrium rate responds to the entire disturbance vector through $(-J)^{-1}F$, mixing C's own diversion gain with its exposure to B's deterioration, and reversal requires the gain to outweigh that exposure.

Corollary 4.3 (Single elasticity)**.** *With $\rho = \eta = \sigma$: $\rho_3^* - \sigma = \sigma(3\alpha_D - 1) + 3(1-\alpha_D)(1-\alpha_T) \geq 0$ whenever $\alpha_D \geq 1/3$, for any α_T (with equality only at $\alpha_D = 1/3$, $\alpha_T = 1$); the reversal is unreachable. At the equal-weights point $\alpha_D = 1/3$, $\alpha_T = 3/4$: $d\log e_{AC}/d\tau = -1/(12\sigma) \to 0^-$.*

This says that the diversion gain cannot offset B's deterioration when domestic and foreign tradables enter a single CES nest (a common elasticity for both margins), given home bias $\alpha_D \geq 1/3$. In fact, since $3\rho_1^* \geq \eta$ whenever $\alpha_D \geq 1/3$, for A to depreciate bilaterally against C requires $\rho > \eta$, i.e., it is necessary (but not sufficient) for the cross-origin elasticity to be greater than the home–import elasticity.

4.4 General symmetric N

With weights $1/(N-1)$ and the $N-2$ bystanders identical (common ν_C), (19)–(20) give

$$d\log P_{M_A} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \qquad d\log P_{M_B} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \qquad d\log P_{M_C} = \frac{\nu_B - 2\nu_C}{N-1}, \tag{33}$$

with $d\log I_A = \frac{m}{N-1}d\tau$, $d\log I_B = \nu_B$, $d\log I_C = \nu_C$. Collecting (24) for B and a representative bystander (rows scaled by $(N-1)/m$):

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \qquad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1, \tag{34}$$

where the counting matrix has determinant N; inverting:

$$\frac{d\log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{N\,D_N} < 0, \qquad \frac{d\log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{N\,D_N}, \qquad \frac{d\log e_{BC}}{d\tau} = \frac{(N-1)\,\rho}{N\,D_N} > 0, \tag{35}$$

which reduce to (32) at $N = 3$ and to (27) at $N = 2$.

Theorem 4.4 (Dilution threshold)**.** *Under* (17)*, the bystander bilateral rate reverses iff*

$$\boxed{\rho > \rho_N^* = N\rho_1^*.} \tag{36}$$

Diverted expenditure is spread across $N-2$ bystanders, each with weight $1/(N-1)$, so stronger cross-origin substitution is required to reverse any individual bilateral rate (dilution).[2] Furthermore, with $\rho = \eta = \sigma$, $\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0$ whenever $\alpha_D \geq 1/N$, so the reversal is again unreachable for single-elasticity models on the home-biased region.

5 Bilateral versus Effective Exchange Rates

Averaging (35) with symmetric weights:

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \boxed{-\frac{\rho_1^*}{D_N}} < 0, \qquad \nu_B - \nu_C = -\frac{(N-1)\rho}{ND_N}\,d\tau. \tag{37}$$

Theorem 5.1 (NEER preservation)**.** *Under symmetry, a unilateral tariff appreciates the tariffing country's NEER for every $N \geq 2$ and every ρ satisfying* (17)*.*

While ρ governs the relative distribution of adjustment across foreign partners, its direct terms cancel from the NEER numerator under symmetry and cannot affect the sign of the adjustment (it nonetheless enters in D_N and therefore still affects the magnitude of the aggregate response). This means that the conventional two-country appreciation result survives multilateralization as a statement on effective exchange rates, robust to N and to changes in ρ.[3]

6 Extensions

6.1 Alternative tariff configurations

Responses are linear in the tariff vector, so configurations are superpositions of directed bilateral shocks (directions a on the same G).

Broad unilateral tariff. A uniform tariff on all $N-1$ partners treats every origin identically, deactivating cross-origin reallocation. Superposing (35) over targets:

$$\frac{d\log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \qquad \forall j: \tag{38}$$

ρ drops out of the numerator, every bilateral rate appreciates equally, and bilateral and effective responses coincide. This shows that bilateral exchange-rate ambiguity arises from discrimination across foreign suppliers, not protection itself.

Symmetric trade war. Adding the mirror shock $dt_{BA} = d\tau$:

$$\left.\frac{d\log e_{AB}}{d\tau}\right|^w = 0, \qquad \left.\frac{d\log e_{AC}}{d\tau}\right|^w = \frac{\rho - \rho_1^*}{D_N}, \qquad \left.\frac{d\nu_A^E}{d\tau}\right|^w = \frac{N-2}{N-1}\,\frac{\rho - \rho_1^*}{D_N}. \tag{39}$$

[2] The exact linearity in N is a symmetric nested-CES result, but it is also true that at any balanced baseline where A and a bystander both import from both foreign origins, sufficiently strong cross-origin substitution still reverses the bystander rate (Appendix B).

[3] Appendix B also shows that the cancellation survives to leading order within a structured asymmetric family but can reverse for a sufficiently under-weighted target, and Appendix C collects the correspondence across N.

Proposition 6.1 (War threshold)**.** *Under* (17)*, in a symmetric war the belligerents' effective rates depreciate, and every bystander appreciates against both, iff* $\rho > \rho_1^*$*, independently of* N*.*

Retaliation cancels the direct bilateral component between A and B, leaving only the bystander reallocation margin. The war threshold is far below the unilateral threshold $N\rho_1^*$ because the direct appreciation against B has been removed. At $\eta = 1$, $\rho_1^* = 1 - m < 1$.

6.2 Real exchange rates

From (10) and (6), $d\log q_{Aj} = \nu_j + \alpha_T(d\log P_{T_j}^{\text{own}} - d\log P_{T_A}^{\text{own}})$; for the unilateral tariff both bilateral real rates take one form at every N:

$$d\log q_{Aj} = \Big(1 - \frac{mN}{N-1}\Big)\,\nu_j \;-\; \frac{m}{N-1}\,d\tau, \qquad j = B, C: \tag{40}$$

attenuated nominal pass-through plus a direct price-index term working toward real appreciation of A. Setting $d\log q_{AC} = 0$ with (35) (for $m < 1/N$):

$$\rho_q^*(N) = \frac{\big[(N-1) - mN\big]\,N\rho_1^* + mN\big[N\alpha_D(\eta-1)+1\big]}{(N-1)(1-mN)} \;>\; N\rho_1^*. \tag{41}$$

Averaging (40):

$$\frac{dq_A^E}{d\tau} = -\Big(1 - \frac{mN}{N-1}\Big)\frac{\rho_1^*}{D_N} - \frac{m}{N-1} \;<\; 0 \qquad \text{for all } \rho, N \ \big(m < \tfrac{N-1}{N}\big). \tag{42}$$

Proposition 6.2 (Real rates)**.** *Under* (17)*: (i) the real bilateral reversal threshold exceeds the nominal one,* $\rho_q^*(N) > N\rho_1^*$*; (ii) the REER appreciates for every* ρ *and* N*,* $dq_A^E/d\tau < 0$*.*

Both parts come from the direct price-index term in (40): the tariff raises A's consumer prices at any nominal rate, which makes real bilateral reversal harder than nominal and adds to the nominal effective appreciation.

7 Relation to the Canonical PTA Model

7.1 Competing exporters

The canonical trade model from the preferential-trade-agreement (PTA) literature has two competing exporters, in the tradition of Viner (1950) and Mundell (1964). Importer A holds only the numeraire y; exporters B, C hold good x; preferences are quasilinear, $U_i = u_i(x_i) + y_i$. A levies specific tariffs t_B, t_C; with q_j the exporter price and p the domestic price,

$$p = q_B + t_B = q_C + t_C, \qquad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \qquad M_A' < 0,\ E_j' > 0, \tag{43}$$

where $M_A(p) = x_A^d(p)$ and $E_j(q_j) = \omega_j - x_j^d(q_j)$ from $u_i' =$ local price. Differentiating in t_B:

$$\frac{dp}{dt_B} = \frac{E_B'}{E_B' + E_C' - M_A'} \in (0,1), \qquad \frac{dq_B}{dt_B} = -\frac{E_C' - M_A'}{E_B' + E_C' - M_A'} < 0, \qquad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \tag{44}$$

Proposition 7.1 (Viner–Mundell)**.** *A discriminatory tariff on B worsens B's and improves C's terms of trade, unconditionally.*

7.2 Relation to our model

An improvement in C's terms of trade is equivalent to a rise in e_{AC} (12). The canonical PTA model delivers it with no additional conditions, while our nested-CES model requires $\rho > 3\rho_1^*$. This difference arises because the PTA environment removes the margins that oppose diversion toward C. Decomposing our threshold:

$$\rho_3^* = 3\rho_1^* = \underbrace{3}_{B\text{–}C \text{ linkage}} + \underbrace{3\alpha_D(\eta - 1)}_{\text{home switching}} - \underbrace{3\alpha_T(1 - \alpha_D)}_{\text{openness offset}}. \tag{45}$$

That is, our model also includes a linkage term that collects the forces running through B–C trade. B's income loss cuts its purchases from C, and relative prices induce C to substitute toward B's cheapened goods. The second term is A's switching toward its own tradable. The third term enters negatively, an openness offset: a larger import share m gives the tariff more foreign expenditure to reallocate, which strengthens diversion and lowers the threshold.

The canonical PTA model deletes the first two opposing forces and the income effects behind the first. What remains is reallocation between the two foreign suppliers, signed directly by $M_A' < 0$ and $E_j' > 0$. This is formally stated as follows.

Proposition 7.2 (Canonical PTA restriction)**.** *The canonical PTA model is the restriction of the three-country model to a subspace on which $\rho_3^* \leq 0$.*

Therefore, the Viner–Mundell result is a direct corollary of our nested-CES model. The canonical restrictions push the threshold down to $\rho_3^* \leq 0$, so the reversal condition $\rho > \rho_3^*$ is always satisfied.

The PTA literature varies in which margins it removes. Competing-exporter models (Bagwell and Staiger, 1999; Ornelas, 2005) impose no B–C trade and quasilinearity, so the sign is unconditional. Bloc and endowment models (Krugman, 1991; Bond and Syropoulos, 1996) instead remove home production of importables; Viner's (1950) identical-goods case is the limit $\rho = \infty$. Kemp and Wan (1976) impose none of these restrictions and claim no sign. Applied GTAP-class and quantitative trade models keep every margin in (45) and have been shown capable of generating either sign in simulations, but they do not study the threshold in analytical terms.

8 Conclusion

Our analysis shows that the general equilibrium response to a tariff is $(-J)^{-1}F$: the tariff creates trade-balance disturbances at unchanged exchange rates, and the adjustment mechanism, regular under the multilateral Marshall–Lerner condition, combines them with nonnegative weights.

At $N = 2$ this reduces to the textbook Marshall–Lerner appreciation result, but at $N \geq 3$, a discriminatory tariff generates foreign-supplier reallocation and hence a mixed-sign disturbance vector. This means that the tariffing country always appreciates against the target, but can depreciate against a bystander if the cross-origin substitution elasticity is sufficiently high.

Despite this ambiguity in bilateral exchange-rate adjustments, aggregation under symmetry shows that A's NEER appreciates for every ρ and N, i.e., the conventional wisdom survives in effective exchange rates. Therefore, a core takeaway from our analysis is the organizing distinction between aggregate *home-versus-foreign adjustment* and *reallocation across foreign suppliers*, where the aggregate margin determines the sign of the effective exchange-rate response, while reallocation determines its bilateral distribution across foreign origins.

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A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$ shares are constant at the weights, $I_A = w_A L_A/[1 - \frac{\tau}{1+\tau}m]$ from (8), and $\mathrm{TB}_A = 0$ reads $e\, m\, w_B L_B = m I_A/(1+\tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1+(1-m)\tau}, \qquad \left.\frac{d\log e}{d\tau}\right|_{\tau=0} = -(1-m) = -\left.\frac{\rho_1^*}{D_2}\right|_{\eta=1}, \tag{46}$$

globally decreasing in τ: the linearized sign in (27) is exact at all tariff levels in this case. ($\eta = 1$ fixes the elasticity, not the weight α_D.)

B Asymmetry

The disturbance-adjustment representation (14) and the multilateral Marshall–Lerner condition (17) do not use symmetry; the exact thresholds of Section 4 and the NEER cancellation of Section 5 do. This appendix establishes what survives without it: the bilateral reversal region survives at any interior baseline, and the NEER result is preserved to leading order within a structured asymmetric family.

General asymmetric baseline. At any balanced baseline, (19)–(24) hold with observed shares (b_{ij}, h_i) and incomes y_i as coefficients. For $N = 3$ at a general baseline (paper appendix), the bystander slope has denominator $\det(-J) > 0$ and numerator quadratic in ρ,

$$\operatorname{sign} \frac{d \log e_{AC}}{d\tau} = \operatorname{sign}\left(c_2\,\rho^2 + c_1\,\rho + c_0\right), \qquad c_2 = b_{AB}\,b_{AC}\,b_{CA}\,b_{CB}\,(1 - h_A)(1 - h_C)\,y_A\,y_C, \quad (47)$$

with c_1, c_0 depending on the full pattern of shares and sizes. Since $c_2 > 0$ whenever A and C each buy from both origins, sufficiently strong cross-origin substitution reverses e_{AC} at any such baseline; the threshold is the largest root, equal to $3\rho_1^*$ at the symmetric point. Away from symmetry no universal bilateral threshold exists.

Structured asymmetry. Let the target carry a multiple κ of the symmetric bilateral share (comparable across N, since the symmetric share shrinks like $1/(N-1)$), with the weight matrix symmetric and balanced:

$$b_{AB} = b_{BA} = \frac{\kappa}{N-1}, \qquad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N-2}, \qquad b_{CC'} = \frac{1 - 2\,b_{CA}}{N-3}. \qquad (48)$$

The reduced system again has two unknowns and solves in closed form. With NEER weights $w_{Aj} = b_{Aj}$, the response is a rational function of (ρ, N, κ) collapsing to $-\rho_1^*/D_N$ at $\kappa = 1$, with numerator $-\kappa\rho_1^*$ times a quadratic $Q(\rho)$ whose leading coefficient is proportional to $(\kappa-1)(N-1-\kappa)$.

Proposition B.1 (Asymmetric NEER)**.** *Holding κ fixed as $N \to \infty$,*

$$N\,\frac{d\nu_A^E}{d\tau} \;\longrightarrow\; -\,\kappa\,\frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} = -\,\kappa \lim_{N\to\infty} \frac{N\rho_1^*}{D_N}. \qquad (49)$$

To leading order, asymmetric exposure scales the symmetric NEER response by the target's relative importance and preserves its sign. At finite N: for $\kappa \geq 1$ all coefficients of Q are positive at large N, so effective appreciation holds for every ρ; for $\kappa < 1$, Q has a single positive root $\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1-\kappa}(1 + O(N^{-1}))$, beyond which the NEER depreciates — an under-weighted target lets depreciations against many bystanders outweigh the appreciation against B, with $\rho_E^* \to N\rho_1^*$ as $\kappa \to 0$. The reversal threshold is of order N. The symmetric NEER result is exact under symmetry and approximately robust within this structured asymmetric family; robustness to arbitrary asymmetric trade networks is not claimed.

C Correspondence across N

object	$N = 2$	$N = 3$	general N
ML condition (17)	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	$-J$ a nonsingular M-matrix
sufficient primitive	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 1)
sign rule	one impact, unconditional	weights $\frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$	sign $\left(\sum_j \vartheta_{ij} F_j\right)$, $\vartheta \geq 0$
bilateral threshold	none (no bystander)	$3\rho_1^*$	$N\rho_1^*$ (symmetric); degree-$(N-1)$ roots in general
real bilateral threshold	none	$\rho_q^*(3)$	$\rho_q^*(N)$ (41)
NEER (unilateral)	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$: sign invariant under symmetry
war threshold	—	ρ_1^*	ρ_1^*, independent of N

D Proofs

Throughout, $m \equiv \alpha_T(1-\alpha_D)$, $P \equiv \rho + \alpha_D(\eta - 1)$, and all derivatives are evaluated at the balanced free-trade baseline of Section 4.1.

Proof of Proposition 3.1. By (19), prices enter (20)–(24) only through differences $\nu_j - \nu_i$, so adding a common constant to every ν_i (including ν_A) leaves each $d\,\mathrm{TB}_i$ unchanged: $\sum_l \widetilde{J}_{il} = 0$ for all i. Summing (9) over i gives $\sum_i \mathrm{TB}_i \equiv 0$ identically, so $\sum_i \widetilde{J}_{il} = 0$ for all l. Under Assumption 1, $\widetilde{J}_{il} \geq 0$ for $l \neq i$, so $\widetilde{J}_{ii} = -\sum_{l \neq i} \widetilde{J}_{il} \leq 0$, and deleting row and column A gives (16), with strict inequality in every row i with $\widetilde{J}_{iA} > 0$.

Thus $-J$ is a Z-matrix (nonpositive off-diagonal entries) whose rows are diagonally dominant with nonnegative diagonal, irreducible, with strict dominance in at least one row. Its diagonal is strictly positive: $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$ with $D = \mathrm{diag}(-J) > 0$ and $B \geq 0$. Row dominance makes every row sum of $D^{-1}B$ at most one, strictly less in some row; with irreducibility, its spectral radius satisfies $r \equiv \rho(D^{-1}B) < 1$. Hence

$$(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k \geq 0} (D^{-1}B)^k D^{-1} \;\geq\; 0.$$

Nonsingularity follows from $r < 1$; $\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0$ because $\det D > 0$ and the eigenvalues $1 - \lambda$ of $I - D^{-1}B$ satisfy $|\lambda| \leq r < 1$, so real eigenvalues are positive and complex ones enter in conjugate pairs with positive products. Finally, the Gershgorin discs of $-J$ are centered at $|J_{ii}| > 0$ with radii at most $|J_{ii}|$, so they lie in the closed right half-plane and touch the imaginary axis only at the origin, which nonsingularity excludes: $\mathrm{Re}\,\lambda(-J) > 0$, i.e., $\mathrm{Re}\,\lambda(J) < 0$. □

Proof of Corollary 3.2. Immediate from (14): $d\nu/d\tau = (-J)^{-1}F$ with $(-J)^{-1} = (\vartheta_{ij}) \geq 0$ elementwise by Proposition 3.1. $\square$

Proof of Proposition 4.1. At $N = 2$, $f_{AB} = f_{BA} = m$ and each country has a single import origin, so $dp_{AB} = d\log P_{M_A} = \nu_B + d\tau$ and $dp_{BA} = d\log P_{M_B} = -\nu_B$; the $(1-\rho)$ term of (22) vanishes and

$$d\log s_{AB} = (1-\eta)\alpha_D(\nu_B+d\tau), \qquad d\log s_{BA} = -(1-\eta)\alpha_D\,\nu_B, \qquad d\log I_A = m\,d\tau, \qquad d\log I_B = \nu_B.$$

Substituting into (24),

$$d\,\mathrm{TB}_A = m\big[d\log s_{BA}+\nu_B\big]-m\big[d\log s_{AB}+m\,d\tau-d\tau\big] = m\big[\big(1+2\alpha_D(\eta-1)\big)\nu_B+\big(1-m+\alpha_D(\eta-1)\big)d\tau\big],$$

which is (26) with $D_2 = 1 + 2\alpha_D(\eta-1)$ and $\rho_1^* = 1 - m + \alpha_D(\eta-1)$; the boxed slope follows. For the elasticity form, hold income fixed and let $M(p)$ denote import volume at consumer price $p = p_{AB}$: import value is $s_{AB}I_A$, so $\varepsilon_M \equiv -\partial\log M/\partial\log p = 1-\partial\log s_{AB}/\partial\log p = 1-(1-\eta)\alpha_D = 1+\alpha_D(\eta-1)$, and $\varepsilon_X = 1+\alpha_D(\eta-1)$ by the mirror computation in B. Differentiating $\mathrm{TB}_A(e,\tau) = X(1/e) - e\,M((1+\tau)e)$ at $\tau = 0$ and balanced trade $X = eM$ gives $\partial\mathrm{TB}_A/\partial e = M(\varepsilon_X+\varepsilon_M-1)$, so $\varepsilon_X+\varepsilon_M-1 = 2\alpha_D(\eta-1)+1 = D_2$ and (17) at $N = 2$ is exactly $\varepsilon_X+\varepsilon_M > 1$. Since $F = m\rho_1^* > 0$ by (25), the slope $-\rho_1^*/D_2$ is negative. $\square$

Proof of Proposition 4.2. From (29), the deviations of each bilateral price from the relevant import index are

$$\begin{aligned}
dp_{AB} - d\log P_{M_A} &= \tfrac{1}{2}(\nu_B - \nu_C + d\tau), & dp_{AC} - d\log P_{M_A} &= \tfrac{1}{2}(\nu_C - \nu_B - d\tau),\\
dp_{BA} - d\log P_{M_B} &= -\tfrac{1}{2}\nu_C, & dp_{BC} - d\log P_{M_B} &= \tfrac{1}{2}\nu_C,\\
dp_{CA} - d\log P_{M_C} &= -\tfrac{1}{2}\nu_B, & dp_{CB} - d\log P_{M_C} &= \tfrac{1}{2}\nu_B,
\end{aligned}$$

and the share changes follow from (22). With flow values $m/2$, (24) reads

$$\begin{aligned}
\tfrac{2}{m}\,d\,\mathrm{TB}_B &= \big(d\log s_{AB} + \tfrac{m}{2}d\tau - d\tau\big) + \big(d\log s_{CB} + \nu_C\big) - \big(d\log s_{BA} + \nu_B\big) - \big(d\log s_{BC} + \nu_B\big),\\
\tfrac{2}{m}\,d\,\mathrm{TB}_C &= \big(d\log s_{AC} + \tfrac{m}{2}d\tau\big) + \big(d\log s_{BC} + \nu_B\big) - \big(d\log s_{CA} + \nu_C\big) - \big(d\log s_{CB} + \nu_C\big).
\end{aligned}$$

Collecting coefficients is elementary; for illustration, the coefficient of ν_B in $\frac{2}{m}d\,\mathrm{TB}_B$ is

$$\underbrace{\tfrac{1}{2}\big[(1-\rho)+(1-\eta)\alpha_D\big]}_{d\log s_{AB}} + \underbrace{\tfrac{1}{2}\big[(1-\rho)+(1-\eta)\alpha_D\big]}_{d\log s_{CB}} + \underbrace{2(1-\eta)\alpha_D - 2}_{-\,d\log s_{BA}-d\log s_{BC}-2\nu_B} = -\big(3\alpha_D(\eta-1)+\rho+1\big) = -D_3.$$

Carrying out the same collection for the remaining coefficients yields (30); the replication package verifies the full collection symbolically. Inverting, as displayed in (31), gives (32). With $D_3 > 0$: the e_{AB} numerator $-(\rho + 3\rho_1^*)$ is negative and the e_{BC} numerator 2ρ positive for all admissible parameters, while $\mathrm{sign}(d\log e_{AC}/d\tau) = \mathrm{sign}(\rho - 3\rho_1^*)$. $\square$

Proof of Corollary 4.3. Substituting $\rho = \eta = \sigma$ into $3\rho_1^* = 3[1 + \alpha_D(\sigma - 1) - \alpha_T(1 - \alpha_D)]$,

$$\rho_3^* - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T).$$

For $\alpha_D \geq 1/3$ the first term is nonnegative and the second is nonnegative, with both zero only at $\alpha_D = 1/3$, $\alpha_T = 1$. Reversal requires the strict inequality $\rho > \rho_3^*$, i.e. $\sigma > \rho_3^*$, which fails whenever $\rho_3^* - \sigma \geq 0$. At $\alpha_D = 1/3$, $\alpha_T = 3/4$: $\rho_3^* - \sigma = \frac{1}{2}$ and $D_3 = 3 \cdot \frac{1}{3}(\sigma - 1) + \sigma + 1 = 2\sigma$, so $d \log e_{AC}/d\tau = (\sigma - \rho_3^*)/(3D_3) = -1/(12\sigma)$. □

Proof of Theorem 4.4. The equilibrium is unique by Proposition 3.1, and the system is invariant under permutations of the $N-2$ bystanders, so the unique solution is permutation-invariant: all bystanders share a common ν_C, and the system reduces to the conditions for B and one representative bystander. From (33),

$$\begin{aligned}
&dp_{AB} - d\log P_{M_A} = \tfrac{N-2}{N-1}(\nu_B + d\tau - \nu_C), && dp_{AC} - d\log P_{M_A} = \tfrac{1}{N-1}(\nu_C - \nu_B - d\tau),\\
&dp_{BA} - d\log P_{M_B} = -\tfrac{N-2}{N-1}\nu_C, && dp_{BC} - d\log P_{M_B} = \tfrac{1}{N-1}\nu_C,\\
&dp_{CA} - d\log P_{M_C} = -\tfrac{1}{N-1}\big[(N-3)\nu_C + \nu_B\big], && dp_{CB} - d\log P_{M_C} = \tfrac{1}{N-1}\big[(N-2)\nu_B - (N-3)\nu_C\big],\\
&dp_{CC'} - d\log P_{M_C} = \tfrac{1}{N-1}(2\nu_C - \nu_B),
\end{aligned}$$

with share changes from (22) and flow values $m/(N-1)$. The two conditions are

$$\begin{aligned}
\tfrac{N-1}{m}\, d\,\mathrm{TB}_B &= \big(d\log s_{AB} + \tfrac{m}{N-1}d\tau - d\tau\big) + (N-2)\big(d\log s_{CB} + \nu_C\big) - \big(d\log s_{BA} + \nu_B\big) - (N-2)\big(d\log s_{BC} + \nu_B\big)\\
\tfrac{N-1}{m}\, d\,\mathrm{TB}_C &= \big(d\log s_{AC} + \tfrac{m}{N-1}d\tau\big) + \big(d\log s_{BC} + \nu_B\big) + (N-3)\big(d\log s_{C'C} + \nu_C\big)\\
&\quad - \big(d\log s_{CA} + \nu_C\big) - \big(d\log s_{CB} + \nu_C\big) - (N-3)\big(d\log s_{CC'} + \nu_C\big).
\end{aligned}$$

Collecting coefficients exactly as in the proof of Proposition 4.2 yields (34); the collection is verified symbolically in the replication package. The counting matrix has determinant $2(N-1) - (N-2) = N$, so inverting gives (35), which contain (32) at $N = 3$ and (27) at $N = 2$. Under (17), $D_N > 0$: the e_{AB} numerator $-[N\rho_1^* + (N-2)\rho]$ is negative and the e_{BC} numerator positive, while $\operatorname{sign}(d\log e_{AC}/d\tau) = \operatorname{sign}(\rho - N\rho_1^*)$. □

Proof of Theorem 5.1. With symmetric weights $w_{Aj} = 1/(N-1)$, using (35),

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \frac{-\big[N\rho_1^* + (N-2)\rho\big] + (N-2)\big[\rho - N\rho_1^*\big]}{(N-1)\,N\,D_N} = \frac{-(N-1)\,N\rho_1^*}{(N-1)\,N\,D_N} = -\,\frac{\rho_1^*}{D_N},$$

the ρ terms cancelling identically. By (25), $\rho_1^* = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T) > 0$ for $\alpha_D, \eta > 0$ and $\alpha_T \leq 1$, and $D_N > 0$ under (17), so the response is negative for every $N \geq 2$ and every admissible ρ. □

Proof of Proposition 6.1. Let primes denote responses to the mirror shock $dt_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in (35) gives $d\log e'_{BA} = -[N\rho_1^* + (N-2)\rho]/(ND_N)$ and $d\log e'_{BC} = (\rho - N\rho_1^*)/(ND_N)$, hence, using $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$,

$$d\log e'_{AB} = \frac{N\rho_1^* + (N-2)\rho}{ND_N}, \qquad d\log e'_{AC} = d\log e'_{AB} + d\log e'_{BC} = \frac{(N-1)\rho}{ND_N}.$$

By linearity, the war response is the sum of (35) and the primed responses:

$$d\log e_{AB}\big|^w = 0, \qquad d\log e_{AC}\big|^w = \frac{\rho - N\rho_1^*}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho-\rho_1^*}{D_N},$$

and by the $A \leftrightarrow B$ symmetry of the war, $d\log e_{BC}|^w = d\log e_{AC}|^w$. The war NEER is $\frac{1}{N-1}[0 + (N-2)(\rho-\rho_1^*)/D_N]$. All three statements have the sign of $\rho - \rho_1^*$ given $D_N > 0$, independently of N. □

Proof of Proposition 6.2. From (10) and (6), only tradable prices move, so $d\log q_{Aj} = \nu_j + \alpha_T(d\log P^{\text{own}}_{T_j} - d\log P^{\text{own}}_{T_A})$ with $d\log P^{\text{own}}_{T_i} = (1-\alpha_D)\, d\log P_{M_i}$, the indices in (33) already being in own currency. Then

$$d\log q_{AB} = \nu_B + m\Big[\tfrac{(N-2)\nu_C}{N-1} - \nu_B - \tfrac{\nu_B + d\tau + (N-2)\nu_C}{N-1}\Big] = \Big(1 - \tfrac{mN}{N-1}\Big)\nu_B - \tfrac{m}{N-1}\,d\tau,$$

and the identical computation for $j = C$ gives (40) (the bracket again collapses to $-\frac{N\nu_C + d\tau}{N-1}$). Setting $d\log q_{AC} = 0$ with ν_C from (35):

$$\big[(N-1) - mN\big]\frac{\rho - N\rho_1^*}{N} = m\, D_N(\rho),$$

which is linear in ρ since $D_N(\rho) = N\alpha_D(\eta-1) + (N-2)\rho + 1$; solving gives (41), using $(N-1) - mN - mN(N-2) = (N-1)(1-mN)$. For $m < 1/N$ the left factor $(N-1) - mN$ is positive, and $D_N(\rho_q^*) > 0$ for $\eta \geq 1$, so $\rho_q^* > N\rho_1^*$: part (i). Averaging (40) over j and substituting (37) gives (42); both terms are negative for $m < (N-1)/N$ under (17): part (ii). □

Proof of Proposition 7.1. Displayed in (44): total differentiation of market clearing gives $dp/dt_B \in (0,1)$, and the signs of dq_B/dt_B and dq_C/dt_B follow from $M'_A < 0$ and $E'_j > 0$. □

Proof of Proposition 7.2. Embed the canonical structure in the nested model's parameter space. A produces none of the importable, so the home share within tradables is $\alpha_D = 0$; the numeraire is traded and there is no non-tradable, so $\alpha_T = 1$; quasilinearity removes the income effects otherwise carried by the linkage term. Evaluating (25), $\rho_1^* = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T) = 0$, hence $\rho_3^* = 3\rho_1^* = 0 \leq 0$, and the reversal condition $\rho > \rho_3^*$ holds for every $\rho > 0$. □

Proof of Proposition B.1. With the weights (48) the bystanders remain interchangeable, so by the uniqueness and symmetry argument of Theorem 4.4 the system again reduces to two unknowns (ν_B, ν_C), with flow values $m\beta$, $m\gamma$, $m\xi$ for $\beta = \kappa/(N-1)$, $\gamma = (1-\beta)/(N-2)$, $\xi = (1-2\gamma)/(N-3)$. Write $\nu_B = b/N$ and $\nu_C = c/N$ with $d\tau$ fixed, and expand the two equilibrium conditions in $1/N$

at fixed (κ, ρ). To leading order, $\beta \to \kappa/N$, $(N-2)\gamma \to 1$, and

$$d \log P_{M_A} = \frac{\kappa\, d\tau + c}{N} + O(N^{-2}), \qquad d \log P_{M_B} = \frac{c - b}{N} + O(N^{-2}), \qquad d \log P_{M_C} = O(N^{-2}).$$

Substituting into (22) and the flow sums, and multiplying the B- and C-conditions by N/m and $N(N-2)/m$ respectively, every coefficient converges, and the limit system in (b, c) is

$$P\,(b - c) = -\,\kappa\rho\, d\tau, \qquad P\,(2c - b) = \kappa(\rho - \rho_1^*)\, d\tau, \qquad P \equiv \rho + \alpha_D(\eta - 1) > 0:$$

the first from the collapse of the tariff term $\beta[(1-\rho)-1]d\tau = -\beta\rho\, d\tau$ and the relative share responses between B and the bystanders; the second from A's diversion toward C net of C's exposure to B, whose $d\tau$ coefficient is $\kappa[\rho - 1 + (1-\eta)\alpha_D + m] = \kappa(\rho - \rho_1^*)$. The limit matrix $P \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ has determinant $P^2 > 0$, so the solution of the exact system converges to the solution of the limit system: adding the two equations, $c = -\kappa\rho_1^*\, d\tau/P$. The NEER weights are $w_{AB} = \beta = O(N^{-1})$ and $w_{AC} = \gamma$ with $(N-2)\gamma \to 1$, so

$$N\,\frac{d\nu_A^E}{d\tau} = N\big[\beta\,\nu_B + (N-2)\gamma\,\nu_C\big]\big/d\tau \;\longrightarrow\; c/d\tau = -\,\frac{\kappa\,\rho_1^*}{\rho + \alpha_D(\eta - 1)},$$

which equals $-\kappa \lim_N N\rho_1^*/D_N$ since $D_N/N \to P$. □